GERARDO ALBERTO AMAYA FUENTES

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LinkedIn | **GitHub** | **Portfolio**

PROFESSIONAL SUMMARY

Full Stack Developer with 4 years of experience building digital government platforms in El Salvador, including online permit portals for the Ministry of National Defense, the Fire Department and municipal governments, used by citizens and businesses. Backend-focused with NestJS, Laravel and PostgreSQL on Google Cloud, with hands-on experience in 3-D Secure payments, electronic document signing, AI agents and OWASP-based security testing. Built AI applications with Python, FastAPI and the Claude API (Anthropic), such as a conversational trip planner and a system that turns scanned invoices into accounting data using OCR and LLMs.

PROFESSIONAL EXPERIENCE

Government of El Salvador San Salvador, El Salvador
Full Stack Developer *Apr 2023 - Present*

- **Ministry of National Defense Permit Portal (MDN):** online portal to apply for and track permits for hazardous materials such as explosives. NestJS and PostgreSQL microservices backend with JWT/JWKS authentication, per-case access control, online payments, Cloud Scheduler jobs for applicant notifications and Cloud Logging monitoring.
- **Fire Department Permit Portal (CBES):** portal for businesses to obtain fire-prevention inspection certificates online, including inspections of vehicles transporting hazardous materials. NestJS backend with a permit gateway and per-agency adapters, dynamic appointment scheduling for inspectors, document review with observations, automated deadlines and notifications, electronically signed PDF resolutions, and lookups against the national ID registry (RNPN) and tax authority (Hacienda). Includes a Gemini-powered AI agent that enforces document rules, classifies applicants and drafts user responses.
- **Distrito Digital La Paz Este:** Laravel platform on Google App Engine that digitizes six municipal procedures (tax clearance, signage, road works, minor and major construction, and antennas) across three municipalities, so citizens no longer need to visit city hall. Citizen and role-based back-office portals, national ID validation, online payments via Serfinsa or in-person, resolutions signed with Uanataca and archived in Cloud Storage, configurable per-district fee schedules and Cloud Scheduler tasks.
- **School Escalation System (SECE):** platform to survey and track around 1,400 public schools ahead of school supply package distribution. Route-assigned field monitors record classrooms, sections, shifts, teachers, infrastructure, connectivity and incidents on site, and a dashboard flags schools needing attention by department.
- **Electronic signature:** PBS signing system and integration with the Uanataca One-Shot API for digital document signing; the Optimizer is self-hosted in Docker so documents and personal data never leave the institutional network.
- **Serfinsa payment gateway:** centralized payment broker designed to be consumed by multiple permit portals. 3-D Secure authentication with Cardinal (merchant JWT), asynchronous queued charges with status polling by ID, payment attempt logging in Cloud SQL (MySQL), cardholder validation against RNPN and security headers (HSTS, X-Frame-Options, X-Content-Type-Options); payment flow troubleshooting with Cloud Logging.
- **Security and pentesting:** automated Jest test suites based on the OWASP Top 10 and OWASP API Security Top 10, covering BOLA/IDOR, BFLA (role-based privilege escalation), SQL and NoSQL injection, XSS, broken authentication and JWT tampering, mass assignment, brute force and rate limiting, SSRF, CORS and security headers; resolved JWT/JWKS authentication incidents in production.
- **Chivo Pets:** landing page for Chivo Pets Veterinary Hospital, El Salvador's first public veterinary hospital and a benchmark in Latin America. Responsive site showcasing the hospital's services (consultations, 24/7 emergency care, surgery, clinical lab and hospitalization).

Asociación Conexión San Salvador, El Salvador
Full Stack Developer Intern *Jan 2023 - Apr 2023*

- Training and certification internship in Java, PHP, Laravel, NestJS and databases.

PERSONAL PROJECTS

Smart Report AI | *Python, FastAPI, PostgreSQL + PostGIS, Claude AI, Next.js, Telegram Bot API* *2026*

- Problem: reporting a road hazard (e.g., a pothole or an open manhole) is slow, and duplicate reports overwhelm responders.
- Solution: citizens report through a Telegram bot in under a minute. A FastAPI service stores each report with its geolocation in PostgreSQL + PostGIS and uses Claude AI to interpret it.
- Reports of the same incident are automatically clustered, so dispatchers see a single case with all evidence in a Next.js dashboard.

DocuFlow AI | *Python, FastAPI, PostgreSQL + pgvector, Celery, Claude AI, Next.js* *2026*

- Problem: manually entering paper invoices into accounting systems is slow and error-prone.
- Solution: a pipeline that runs OCR on scanned invoices and uses Claude AI to extract accounting fields. Each field carries a measured confidence score, so human review focuses only where it is needed.

- Processing runs in the background with Celery queues, and semantic search over pgvector answers questions with citations to the source document.

Waypoint AI | *Python, FastAPI, PostgreSQL + PostGIS, Claude AI, Next.js* *2026*

- Natural-language trip planner for El Salvador, e.g., “beach and local food, leaving from San Salvador, at most two hours of driving”.
- Claude AI turns the request into structured constraints, and a deterministic engine builds the itinerary from real places and distances in PostGIS, so the AI cannot hallucinate destinations or routes.

Real Estate Lead Management | *Angular, TypeScript, Node.js, Express, MongoDB, Playwright* *2026*

- Module to capture and track real estate leads, with an Angular frontend and a Node.js/Express REST API.
- Metrics dashboard computed in a single MongoDB aggregation query, plus end-to-end tests with Playwright.

Travel Quote | *Node.js, Express, Sequelize, MySQL, Docker* *2025*

- Quoting system for tourist transportation services, with a layered clean-architecture REST backend, MySQL persistence through the Sequelize ORM and containerized deployment with Docker.

EDUCATION

Universidad Francisco Gavidia San Salvador, El Salvador
B.S. in Computer Science Engineering, coursework completed

Instituto Técnico Ricaldone, *Technical High School Diploma* *2017 - 2019*

CERTIFICATIONS

- **Associate Cloud Engineer**, Google Cloud. Sep 2026, valid through Sep 2029.
- **MTA: Database Fundamentals**, Microsoft. Jan 2022.
- **Scrum Fundamentals Certified**, VMEdu. Credential ID 982714.
- **Mobile App Development**, Google. Jul 2023. Android app development with Kotlin.
- **PL/SQL Table Functions**, Oracle.
- **Google Cloud Skill Badges (8):** Cloud Run, CI/CD, IAM, Vision API, load balancing, VPC and development environments.

Additional Training

- **SQL Server Specialization**, FORMAS BID
- **Oracle AWS (Basic, Intermediate and Advanced)**, INSAFORP
- **Backend Developer Bootcamp**, INSAFORP and Asociación Conexión
- **Advanced Web Programming**, INSAFORP and Asociación Conexión
- **Java Fundamentals**, INSAFORP
- **Object-Oriented Programming with Python**, INSAFORP
- **Data Structures and Functions in Python**, INSAFORP
- **Python Programming**, Presidential Secretariat for Innovation

TECHNICAL SKILLS

Backend: Node.js, NestJS (TypeORM, Prisma), Express, Python (FastAPI, Django, Flask, Celery), PHP (Laravel), Java, C# .NET

Frontend: Next.js, React, Angular, Vue.js, TypeScript, Tailwind CSS

Mobile: Kotlin (Android)

Databases: PostgreSQL (PostGIS, pgvector), MySQL, SQL Server (T-SQL), Oracle (PL/SQL), MongoDB

Cloud and DevOps: Google Cloud (GKE, App Engine, Cloud Run, Cloud Scheduler, Cloud SQL, Cloud Storage, Pub/Sub, Secret Manager, IAM, Cloud Logging), Docker, CI/CD

Security and QA: Web and API pentesting (OWASP Top 10 and API Security Top 10), 3-D Secure, JWT/JWKS, Jest, Playwright, Swagger

Artificial Intelligence: Claude API (Anthropic), Gemini, OpenAI-compatible APIs, AI agents, OCR and LLM-based extraction, semantic search

Languages: Spanish (native), English (intermediate)

REFERENCES

- **Ivan Orlando Moran Interiano**, Government of El Salvador. +503 6028-8419, ivan.moran@goes.gob.sv
- **Gilberto Lara**, Asociación Conexión. +503 2249-2307, galara@conexion.sv